UNIVERSITY OF RWANDA
COLLEGE OF SCIENCE AND TECHNOLOGY
SCHOOL OF ICT
DEPARTMENT OF COMPUTER AND SOFTWARE ENGINEERING
ACADEMIC YEAR: 2025-2026
DATE: On Monday, April 06th, 2026

MODULE: MOBILE APPLICATION SYSTEMS AND DESIGN

REG. NUMBER: 223001019

LAB 6: PUSH NOTIFICATIONS WITH FIREBASE CLOUD MESSAGING (FCM)

⊛ OBJECTIVES

When the app was open (foreground), and the notification arrived instantly, and the two ~~thing~~ things happened, an overlay banner appeared at the top of the screen and a popup dialog showed the notification title and the body. The message was also saved in the Alerts tab history inside the app.

When the app was in the background, the notifications was delivered by the Android system directly and appeared in the notification drawer at the top of the screen. Tapping it opened the app automatically.

When the app was closed (terminated), I could still hear the notification sound and see it in the notification bar. Tapping it launched the app from scratch.

The device token was displayed on the Token screen inside the app and I used it to send a test message from Firebase console successfully.

⊛ Key steps followed:

1. Created a Flutter project using flutter create whatsapp_fcm
2. Created a Firebase project on console.firebase.google.com and registered the Android app with the package name: "com.example.whatsapp_fcm".
3. Downloaded google-services.json and placed it inside "android/app/".
4. Ran "flutterfire configure" to generate "firebase_options.dart" with the project credentials.

5. Added firebase_core, firebase_messaging and flutter_local_notifications to the pubspec.yaml.

6. Enable core library desugaring in "build.gradle.kts" by setting "isCoreLibraryDesugaringEnabled = true" and upgrading desugar_jdk_libs to version 2.1.4.

7. Added "POST_NOTIFICATIONS" permission and FCM metadata inside the "<application>" tag in "AndroidManifest.xml".

8. Created a notification channel in "MainActivity.kt with "IMPORTANCE_HIGH" so background notifications display correctly on Android 8.0 and above.

9. Built an FCMService singleton in Dart to request permission, fetch the device token and listen for incoming messages.

10. Retrieved the device token using FirebaseMessaging.instance.getToken() and displayed it on a dedicated screen

11. Opened Firebase console, went to messaging, created a new campaign, pasted the device token and sent a test notification.

12. Verified the notification was received while the app was open, in the background and closed.

Challenges faced:

- Got a build error about core library desugaring, fixed by upgrading desugar_jdk_libs from 2.0.4 to 2.1.4.
- Notifications were not showing when the app was closed, fixed by creating the notification channel in MainActivity.kt at the native android level.
- Got an overflow error in the notification popup, fixed by constraining the dialog width and adding "TextOverflow.ellipsis" to text widgets.
- The "AndroidManifest.xml" had spaces inside attribute values which caused resource linking to fail, fixed by rewriting the file cleanly.

Conclusion

Firebase Cloud Messaging was successfully integrated into the Flutter app. Push notifications were received and displayed ~~Gott~~ correctly in all three app states; foreground, background and terminated. The device token was used to target the specific device from Firebase console.

CamScanner

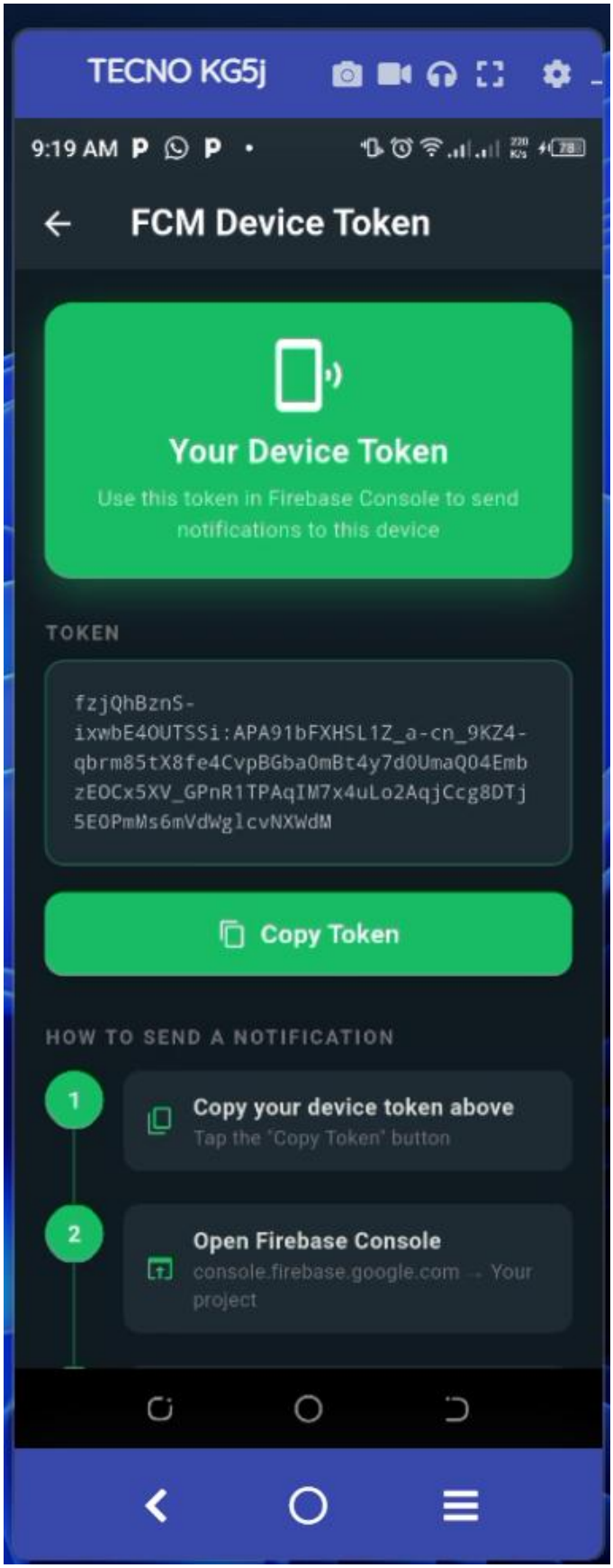

Figure 1: FCM Device Token Screen

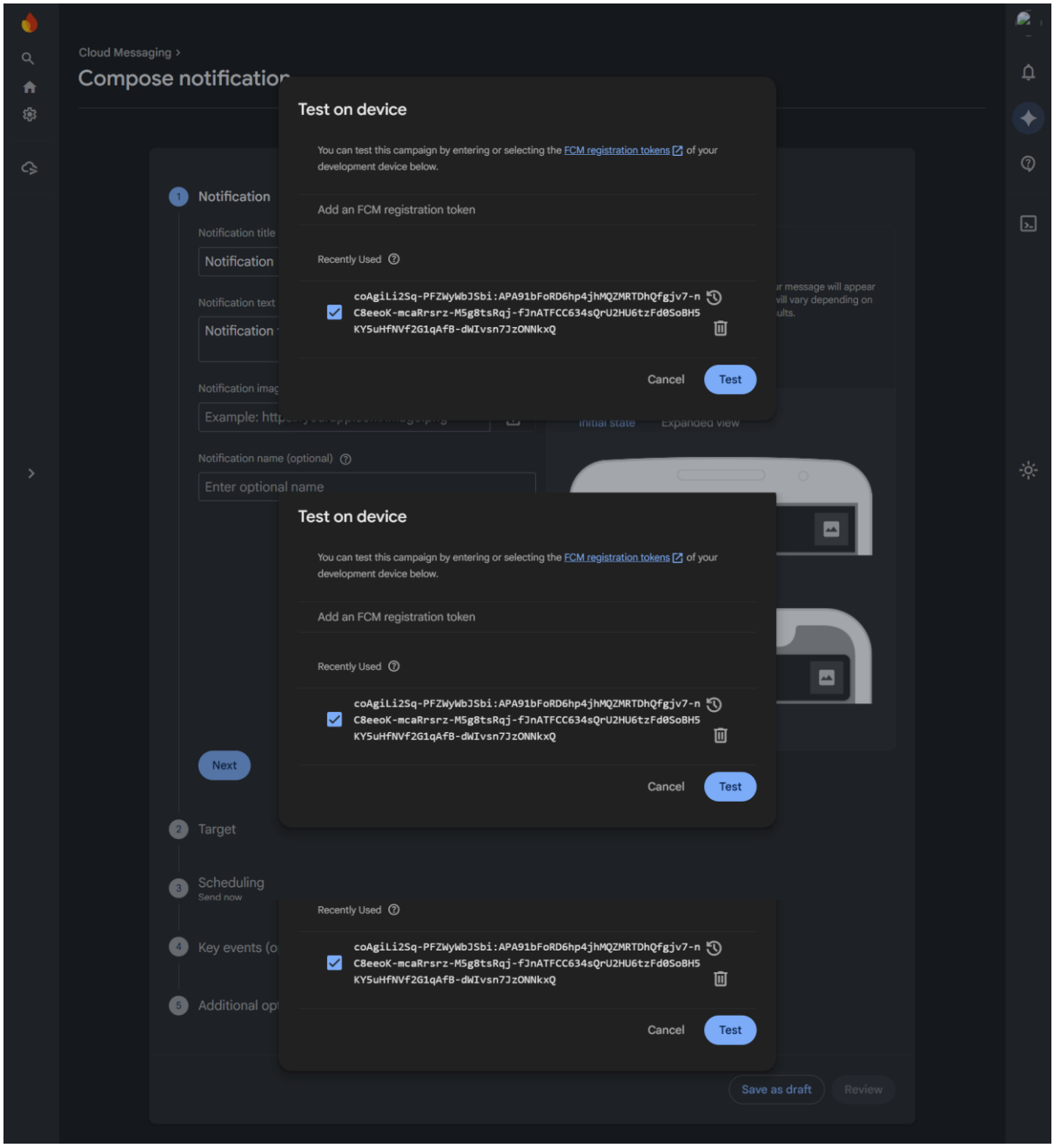

Figure 2: Firebase console - Test

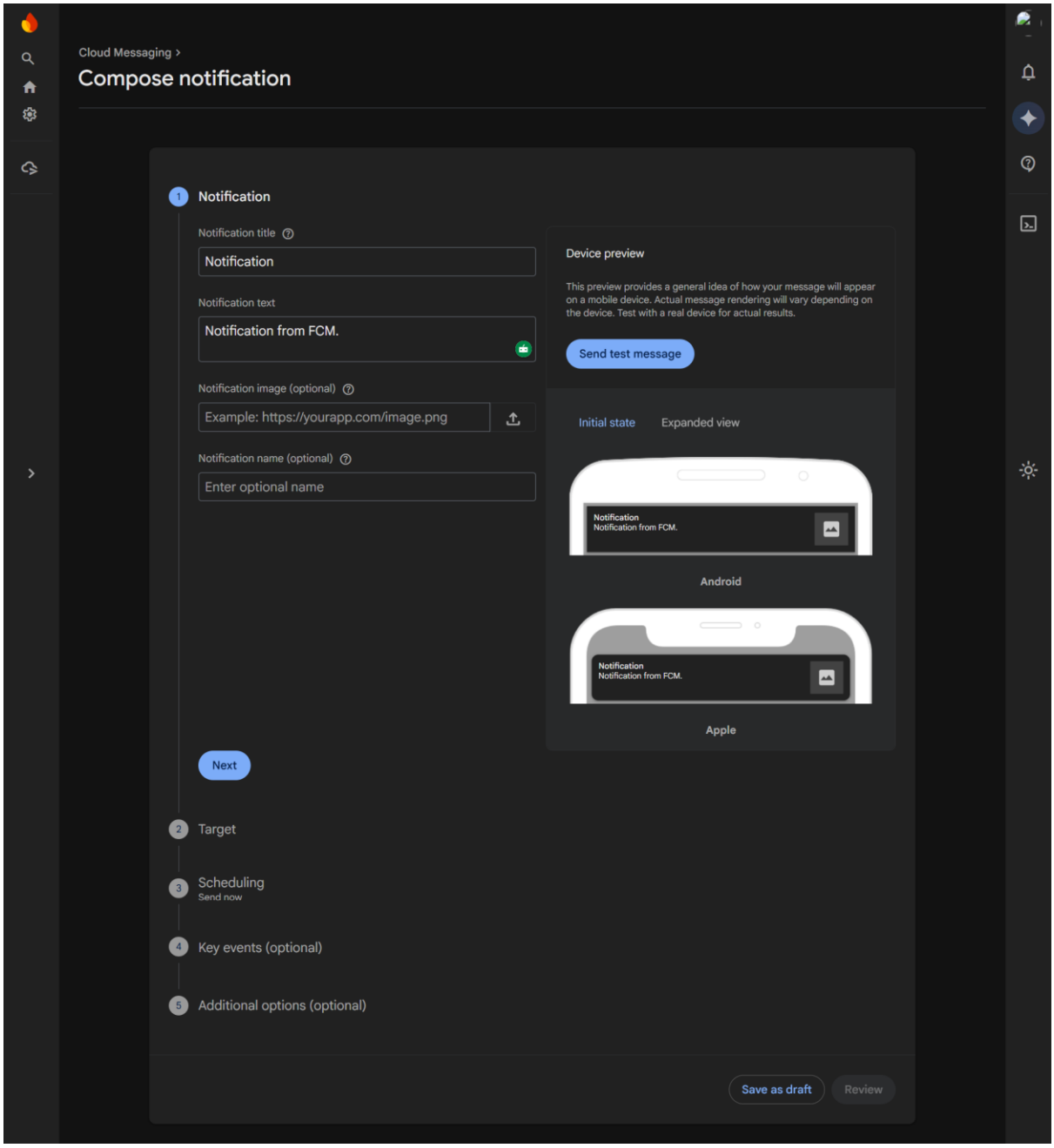

Figure 3: Firebase console

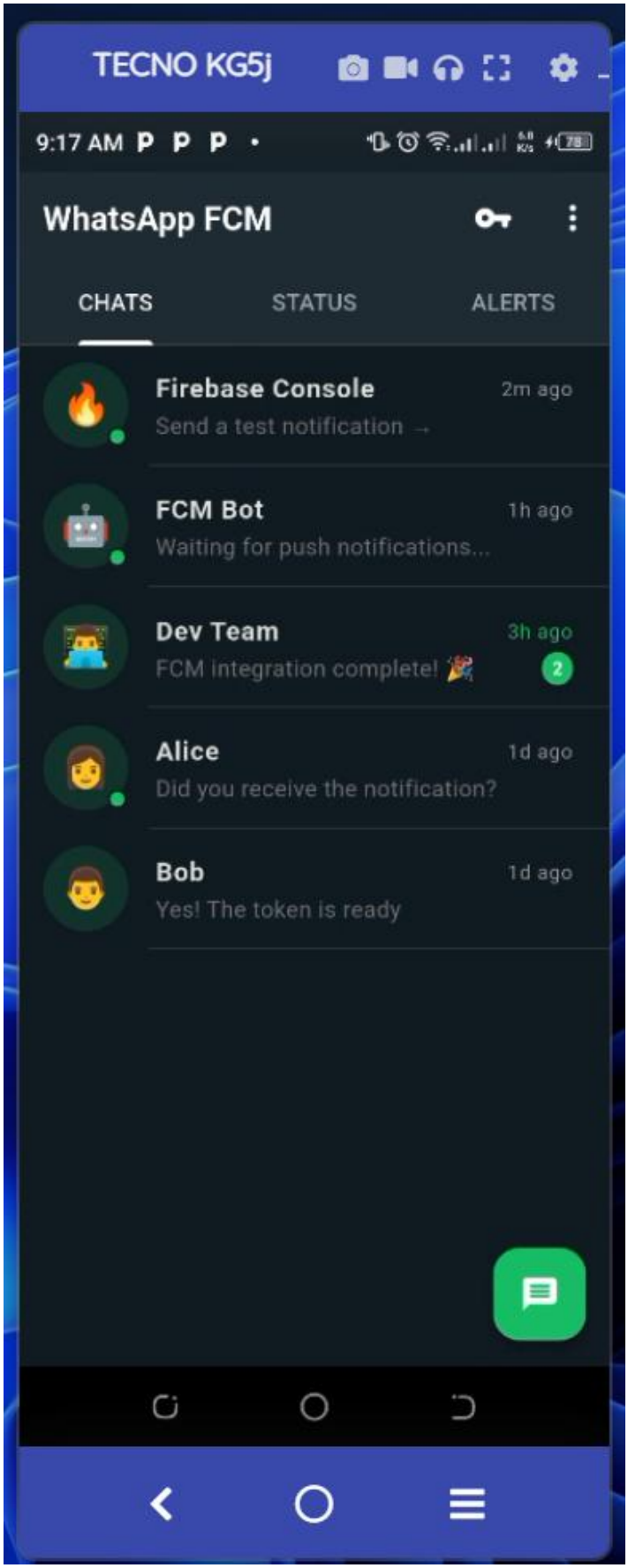

Figure 4: Home Page

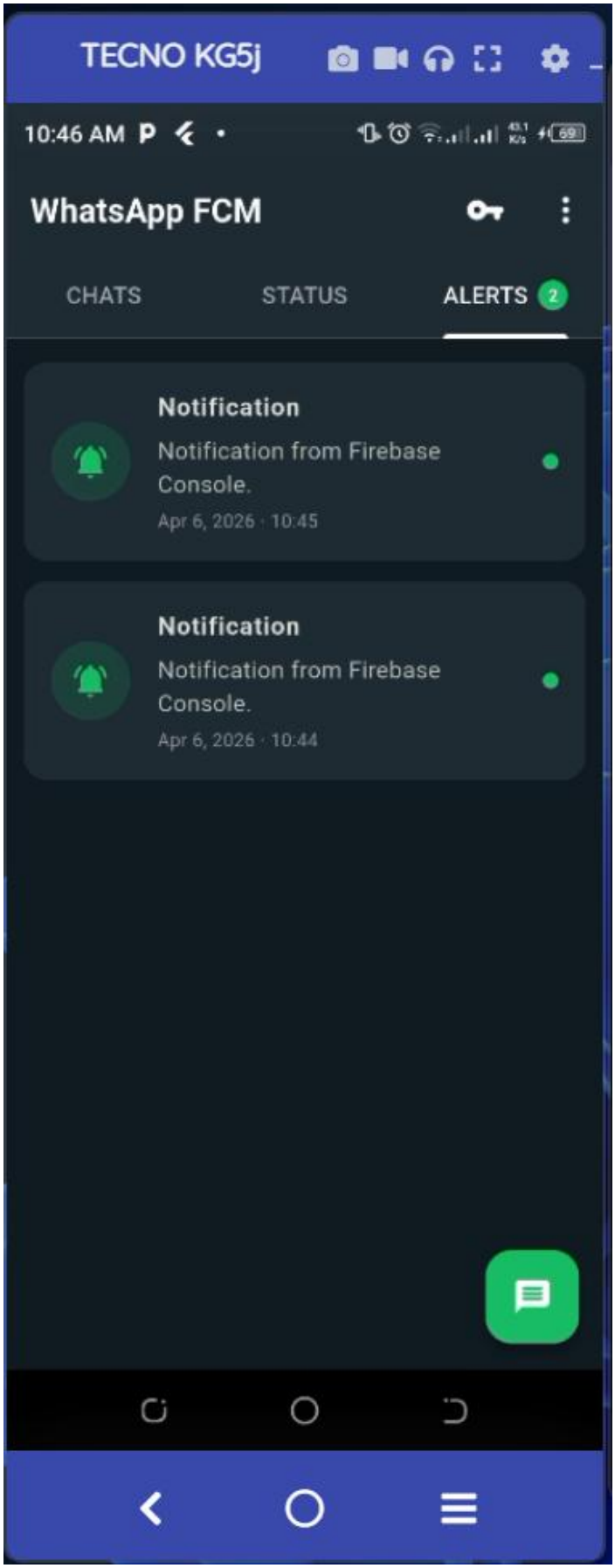

Figure 5: Notification history

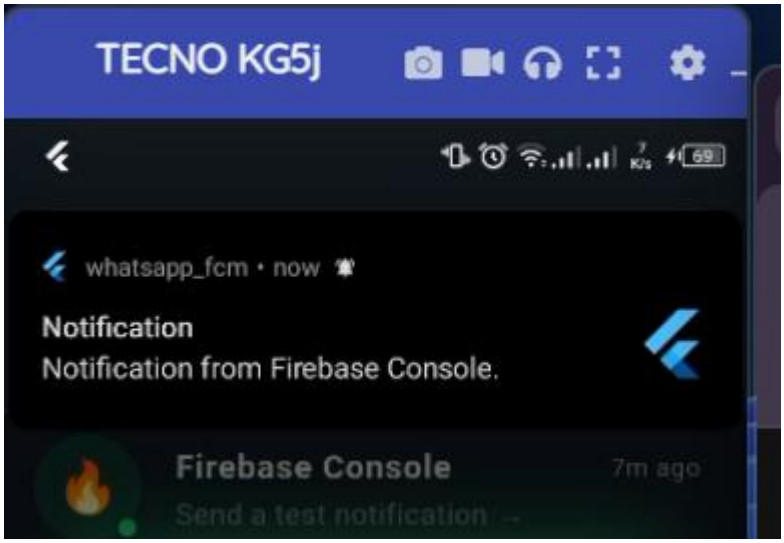

Figure 6: Notification-1-1

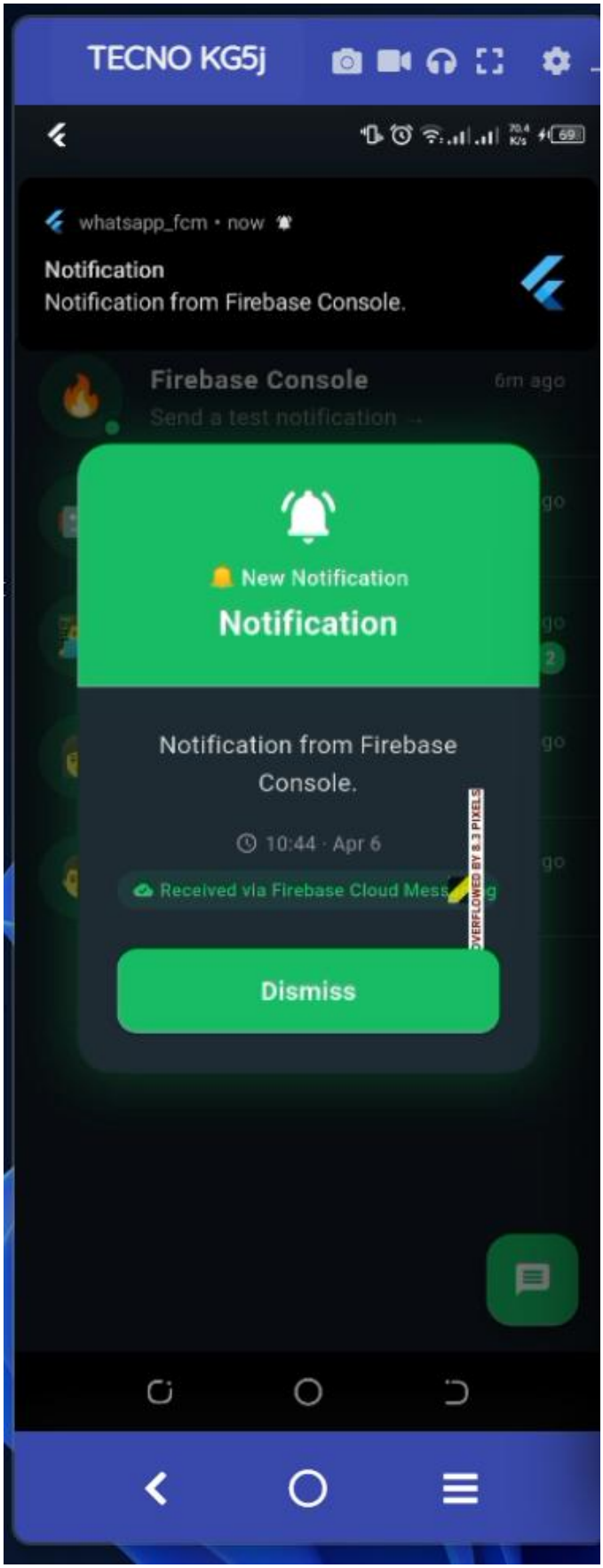

Figure 7: Notification-1

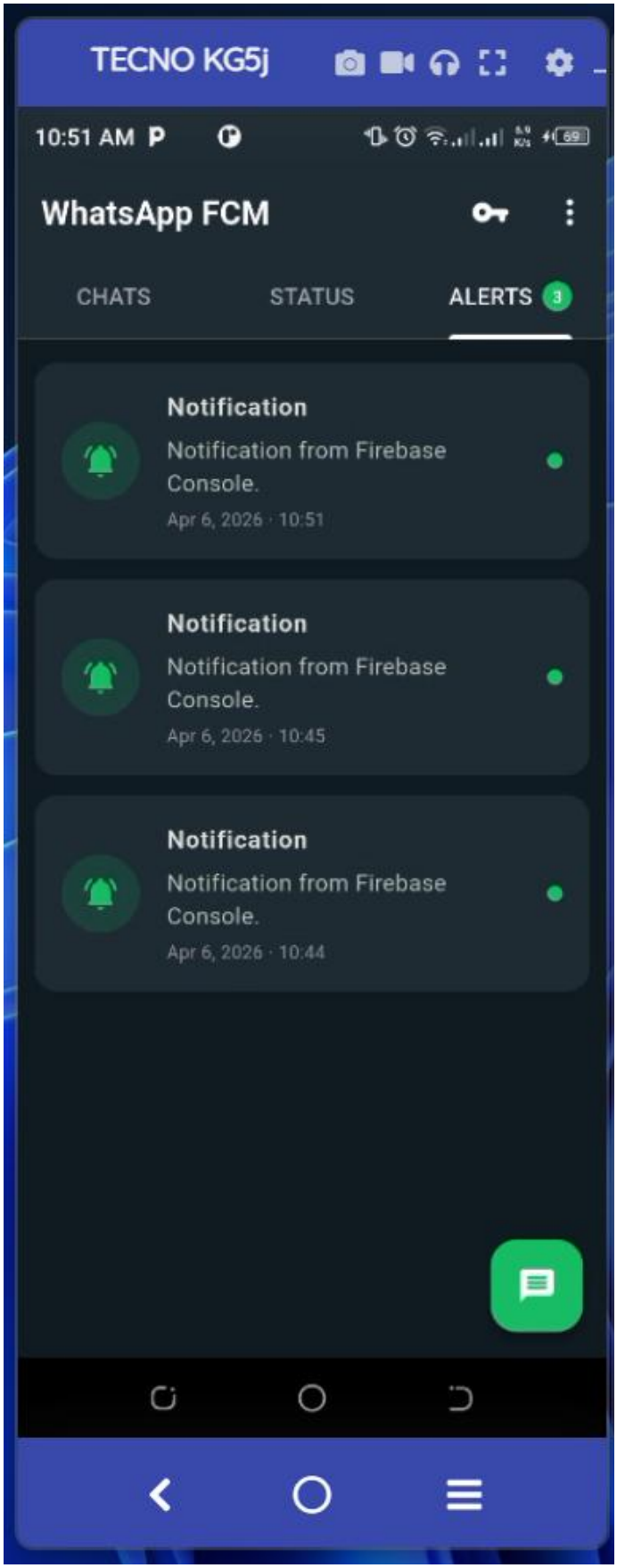

Figure 8: Test-notification-1-app-closed-1-onNotification tap

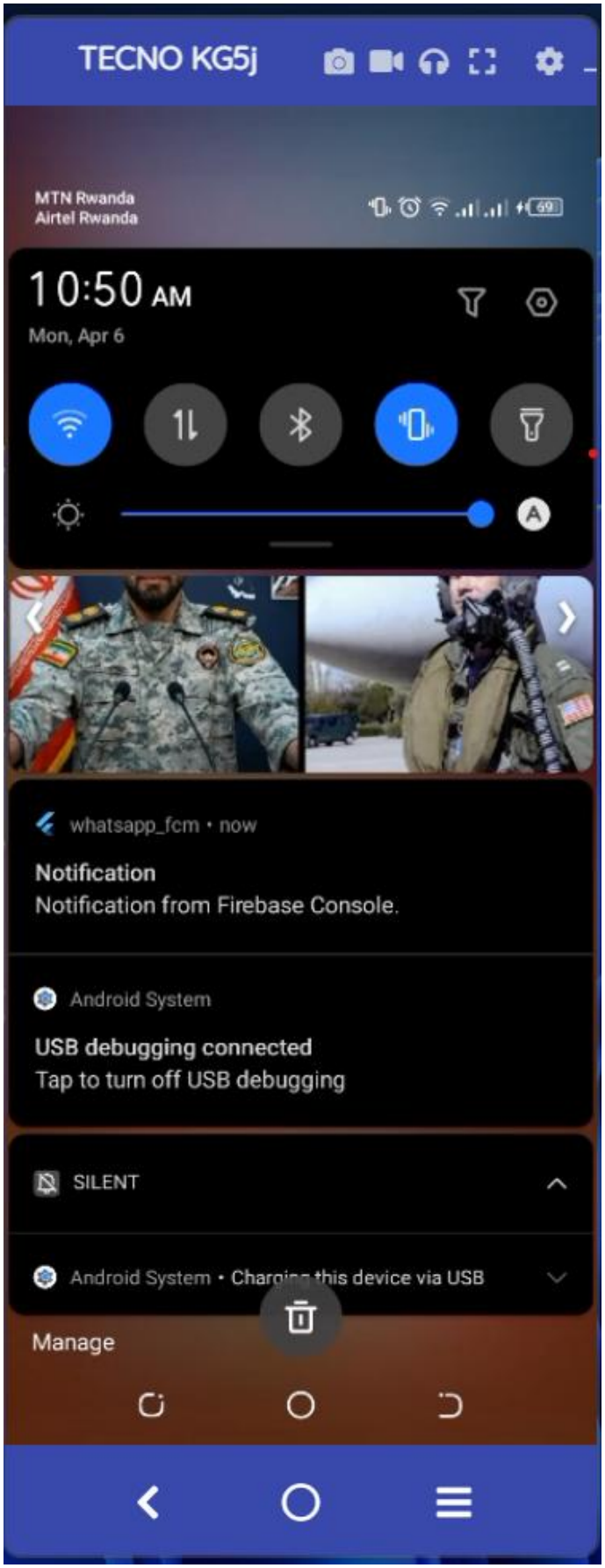

Figure 9: Test-notification-1-app-closed-1

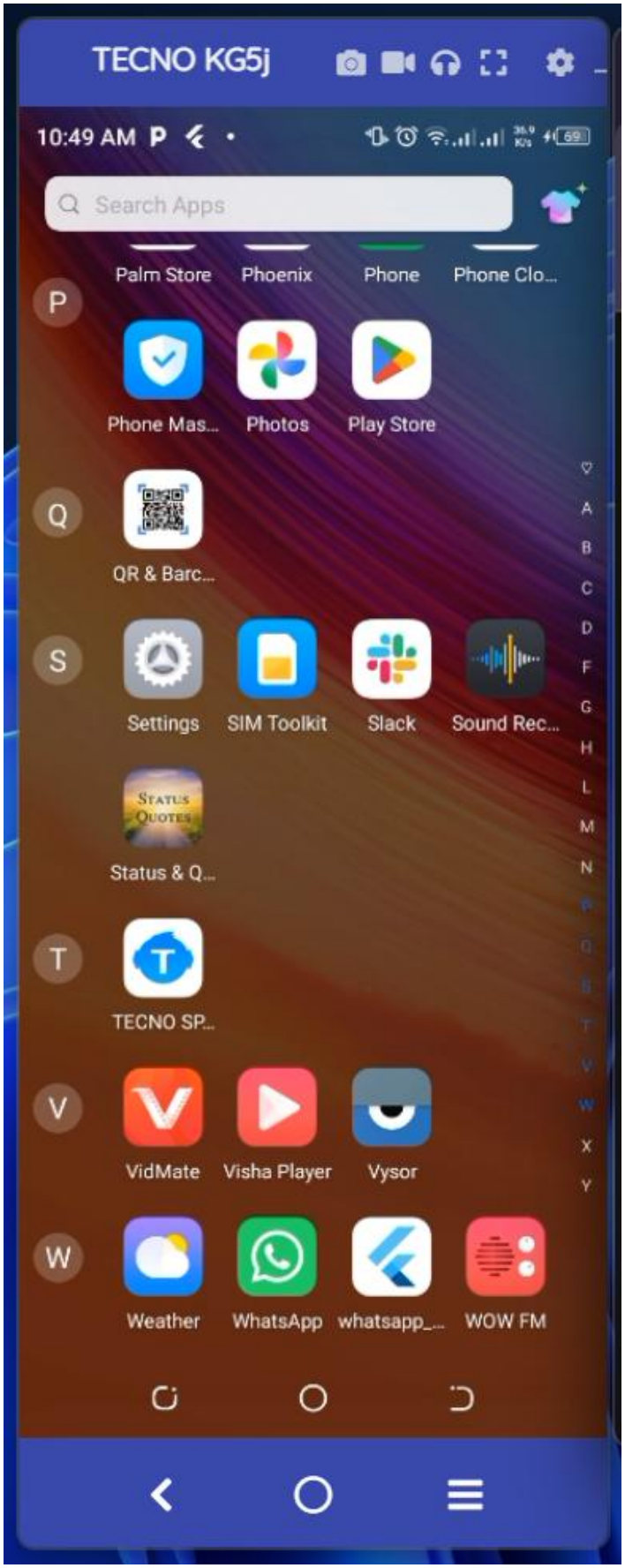

Figure 10: Test-notification-1-app-closed

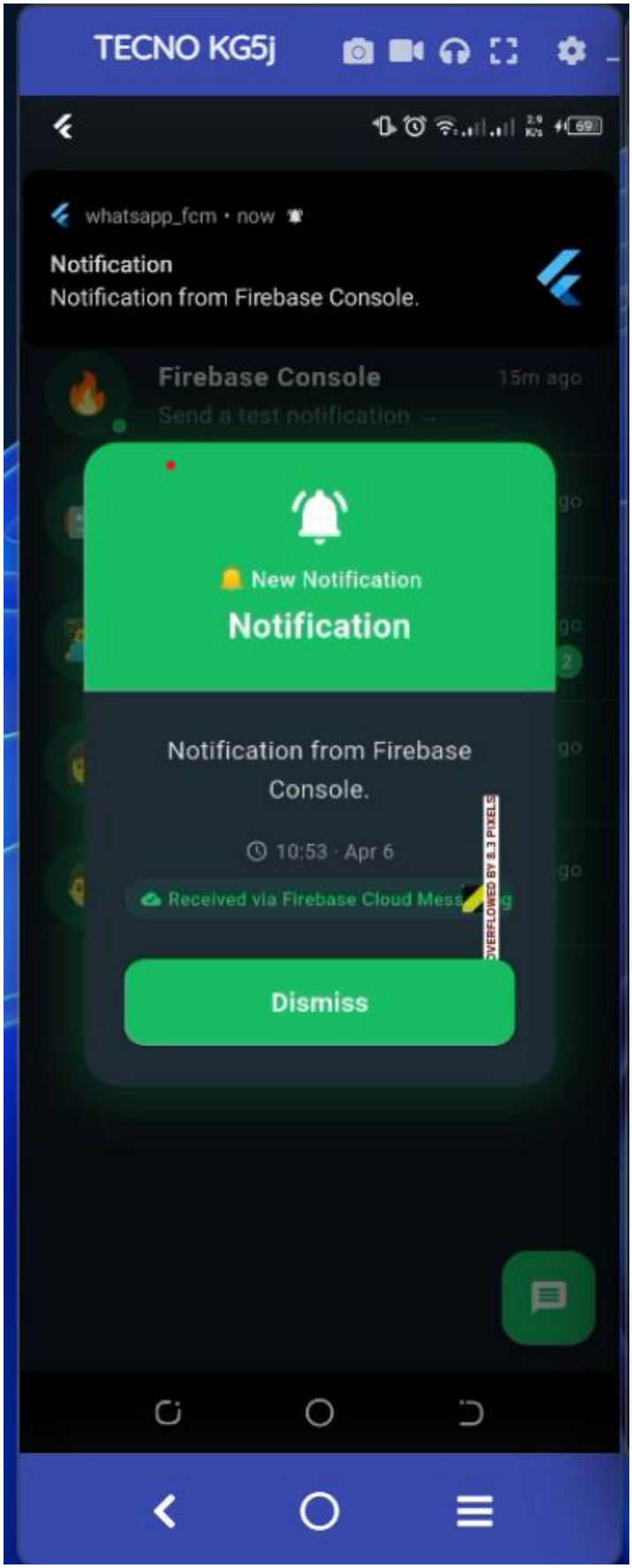

Figure 11: Test-notification-1-app-open - Notified

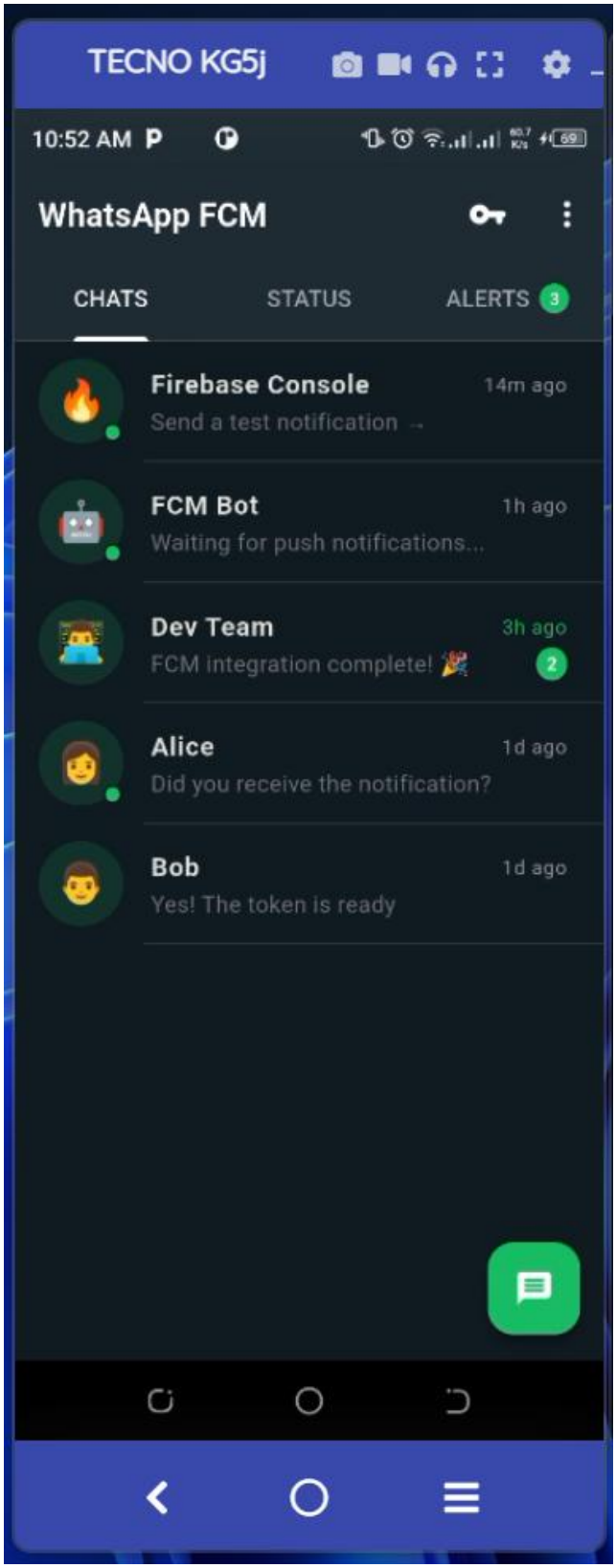

Figure 12: Test-notification-1-app-open

Figure 13: Test-notification-1

The overflow error in the pop-ups, are all fixed and shown within the demo video found in the repo:
https://github.com/SilasHakuzwimana/whatsapp_fcm_lab/tree/88cdc504792fe51c4b111b88ef0fd84070f453b3/demo_video